



10361 - What Makes Early Galaxies Shine So Bright? -- Insights from Ultra-Deep Spectroscopy of MoM-z14 at $z=14.44$

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Rohan Naidu (PI)	Massachusetts Institute of Technology
Prof. Pascal Oesch (CoI) (ESA Member) (CoPI)	University of Geneva, Department of Astronomy
Prof. Erica Nelson (CoI)	University of Colorado at Boulder
Dr. Joel Leja (CoI)	The Pennsylvania State University
Prof. Pratika Dayal (CoI) (CSA Member)	Canadian Institute for Theoretical Astrophysics
Dr. Kasper Elm Heintz (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Darach Watson (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Fabian Walter (CoI) (ESA Member)	Max Planck Institute for Astronomy
Prof. Akio Inoue (CoI)	Waseda University
Dr. Mauro Stefanon (CoI) (ESA Member)	Universitat de Valencia
Prof. Robert Andrew Simcoe (CoI)	Massachusetts Institute of Technology
Prof. Garth D. Illingworth (CoI)	University of California - Santa Cruz
Dr. Jorrit Matthee (CoI) (ESA Member)	Institute of Science and Technology Austria
Dr. Sandro Tacchella (CoI) (ESA Member)	University of Cambridge
Mr. Andrea Weibel (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Dr. Yuma Sugahara (CoI)	Waseda University
Ecaterina Leonova (CoI) (ESA Member)	Universiteit van Amsterdam
Dr. Roberto Decarli (CoI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Dr. Katherine Suess (CoI)	University of Colorado at Boulder
Prof. Sune Toft (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Rychard Bouwens (CoI) (ESA Member)	Universiteit Leiden

JWST Proposal 10361 (Created: Friday, March 13, 2026, 2:06:42PM Eastern Standard Time) - Overview

<i>Name</i>	<i>Institution</i>
Prof. John Chisholm (CoI)	University of Texas at Austin
Prof. Ivo Labbe (CoI)	Swinburne University of Technology
Prof. Arjen van der Wel (CoI) (ESA Member)	Universiteit Gent
Dr. John R. Weaver (CoI)	Massachusetts Institute of Technology
Dr. Katherine E. Whitaker (CoI)	University of Massachusetts - Amherst
Mr. Daniel K. Magee (CoI)	University of California - Santa Cruz
Dr. Benjamin D. Johnson (CoI)	Harvard University
Dr. Sownak Bose (CoI)	Harvard University
Prof. Sirio Belli (CoI) (ESA Member)	Universita di Bologna
Alba Covelo-Paz (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Prof. Pieter van Dokkum (CoI)	Yale University
Dr. Gabriel Brammer (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Yoshinobu Fudamoto (CoI)	Chiba University
Dr. Hakim Atek (CoI) (ESA Member)	CNRS, Institut d'Astrophysique de Paris
Dr. Guido Roberts-Borsani (CoI) (ESA Member)	University College London
Dr. Romain Alexis Meyer (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Emma Giovinazzo (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Dr. Yijia Li (CoI)	Northwestern University
Ms. Clara Lucie Pollock (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Callum Witten (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Prof. Risa H. Wechsler (CoI)	Stanford University
Dr. Julian B. Munoz (CoI) (Contact)	University of Texas at Austin
Dr. Anna G de Graaff (CoI)	Center for Astrophysics Harvard & Smithsonian

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	mom-cos-g235m-5s-3d-v2	NIRSpec MultiObject Spectroscopy	(2) mom-z14-cat-small

ABSTRACT

JWST Proposal 10361 (Created: Friday, March 13, 2026, 2:06:42PM Eastern Standard Time) - Overview

JWST has revealed a stunning population of luminous galaxies at $z > 10$ whose remarkable brightness continues to confound the community. Here we propose extremely deep (24h) R=1000 G235M spectroscopy of the most distant such galaxy, MoM-z14 at $z=14.44$. MoM-z14's prism spectrum shows tantalizing hints of *six* UV lines from *four* chemical elements, making it an exceptionally promising target to unravel the bright galaxy problem. Intriguingly, this source also shows an almost non-existent damping wing despite lying at an epoch where virtually every reionization model expects this galaxy to be blanketed by neutral gas.

With this program we will:

- (i) Test what exactly is powering this source --- e.g., is there an unusually high gas density consistent with extreme starbursts or the broad-line region of an AGN? Does the ionizing spectrum imply a top-heavy IMF or accretion onto a black hole?
- (ii) Derive robust chemical abundances ([N/C], [N/O], [C/O]) to test the spectacular suggestion that MoM-z14, globular clusters, and ancient Milky Way stars share a unique abundance pattern, which is thought to arise in dense clusters that may host exotic objects such as supermassive stars.
- (iii) Detect Lyman-alpha if it exists, confirming the earliest known ionized bubble, perhaps supporting hints from the cosmology community that a modestly ionized universe at $z > 10$ ($\sim 10\%$) may defuse some of the recent challenges to LCDM.

Forging connections from the Milky Way to the cosmic frontier to the ultimate fate of the universe, this program will have remarkable ``out-of-field" impact by further targeting ~ 300 spectroscopically confirmed galaxies.

OBSERVING DESCRIPTION

We propose ultra-deep NIRSpec MSA observations targeting the most distant confirmed galaxy MoM-z14 using the G235M grating.

Our main goal is to detect the faint UV emission lines that are seen at low significance in the existing prism spectrum of this source. Hence, we will use the G235M/F170LP grating/filter combination to confirm these lines, and deblend them, for significantly improved measurements of the physical properties of this source. The required limiting line flux is $1.5e-19$ erg/s/cm², which can be reached in 24 hours of on source integration.

In our MSA mask design, we will use a "normal" 3slitlet dither pattern, but will assign 5 slitlets to the main target for improved background subtraction. Our exposures are split into 7 MSA masks, in which we will switch out filler targets while observing the main source in every mask. The

JWST Proposal 10361 (Created: Friday, March 13, 2026, 2:06:42PM Eastern Standard Time) - Overview

filler targets are drawn from the extensive (low-resolution or WFSS) spectroscopic sample in this field (e.g., CAPERS, MoM, COSMOS-3D), focusing on $z > 6$ galaxies as well as LRDs. Each filler source will receive exposure times of 4-24h, depending on its brightness.

Using the ETC, we find that the SNR for our observations changes by $\sim 7\%$ between low and high background. We therefore request moderately background limited observations and have set the requirement to a background no more than the 30th percentile above the minimum.

Proposal 10361 - Targets - What Makes Early Galaxies Shine So Bright? -- Insights from Ultra-Deep Spectroscopy of MoM-z14 at z=1...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	mom-z14-cat-small	RA: 10 00 26.4582 (150.1102425d) Dec: +02 20 37.04 (2.34362d) Equinox: J2000		
Comments: Description=[]					

Proposal 10361 - Observation 1 - What Makes Early Galaxies Shine So Bright? -- Insights from Ultra-Deep Spectroscopy of MoM-z14 ...

Observation	Proposal 10361, Observation 1: mom-cos-g235m-5s-3d-v2											Fri Mar 13 19:06:42 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(2)	mom-z14-cat-small			RA: 10 00 26.4582 (150.1102425d) Dec: +02 20 37.04 (2.34362d) Equinox: J2000							
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCам Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: After Target ACQ											
	Science Aperture: MSA Center											
	Primary Candidate List: top (2 sources)											
	Filler Candidate List: mom-z14-cat-small (1000 sources)											
Reference Stars	Spectral Overlap Map: jwst-nirspec-g235m											
	Spectral Overlap Threshold: 1.5											
Dithers	#	Dither Type										
	1	NONE										

Proposal 10361 - Observation 1 - What Makes Early Galaxies Shine So Bright? -- Insights from Ultra-Deep Spectroscopy of MoM-z14 ...

Confirmation	NIRSpec MultiObject Spectroscopy		Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time			
	1		After Target Acq	NRSIRS2	5	1	1	379.311			
	2		After Target Acq	NRSIRS2	5	1	1	379.311			
Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1	5 Shutter Slitlet With Gaps	150.090833333333 334 Degrees 2.30063888888888 89 Degrees	246.51200878905 982			3	6	12342.201
	2	1 (G235M/F170LP)	c1	5 Shutter Slitlet With Gaps	150.090833333333 334 Degrees 2.30063888888888 89 Degrees	246.51200878905 982			3	6	12342.201
	3	1 (G235M/F170LP)	c1	5 Shutter Slitlet With Gaps	150.090833333333 334 Degrees 2.30063888888888 89 Degrees	246.51200878905 982			3	6	12342.201
	4	1 (G235M/F170LP)	c1	5 Shutter Slitlet With Gaps	150.090833333333 334 Degrees 2.30063888888888 89 Degrees	246.51200878905 982			3	6	12342.201
	5	1 (G235M/F170LP)	c1	5 Shutter Slitlet With Gaps	150.090833333333 334 Degrees 2.30063888888888 89 Degrees	246.51200878905 982			3	6	12342.201
	6	1 (G235M/F170LP)	c1	5 Shutter Slitlet With Gaps	150.090833333333 334 Degrees 2.30063888888888 89 Degrees	246.51200878905 982			3	6	12342.201
	7	1 (G235M/F170LP)	c1	5 Shutter Slitlet With Gaps	150.090833333333 334 Degrees 2.30063888888888 89 Degrees	246.51200878905 982			3	6	12342.201
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F250M	DEEP8	10	2	6	3	12143.287		
	2	F200W	F277W	DEEP8	10	2	6	3	12143.287		
	3	F150W	F250M	DEEP8	10	2	6	3	12143.287		
	4	F200W	F277W	DEEP8	10	2	6	3	12143.287		
	5	F150W	F250M	DEEP8	10	2	6	3	12143.287		
	6	F200W	F277W	DEEP8	10	2	6	3	12143.287		
	7	F150W	F250M	DEEP8	10	2	6	3	12143.287		

Proposal 10361 - Observation 1 - What Makes Early Galaxies Shine So Bright? -- Insights from Ultra-Deep Spectroscopy of MoM-z14 ...

Special Requirements	Group Visits within 53.0 Days Visits Same PA No Parallel Attachments Background Limited. Background no more than 30th percentile above minimum MSA Planned Aperture PA 246.5127 to 246.5127 Degrees (V3 107.9381303 to 107.9381303)
----------------------	---